Working with Vectors

Exam code: 9MA0

Download PDF

Written by: Amber
Updated on 16 February 2026

SaveMyExams

Did this video help you? Yes No

Working with vectors

Help me understand

Vectors are used throughout mechanics to describe forces and motion

How are vectors used in Mechanics?

- **Vector** questions are often embedded in a Mechanics context
- Vectors will most commonly represent **forces**, **accelerations** or **velocities**, they can also represent displacement
- Newton's Second Law $\mathbf{F} = m\mathbf{a}$ is essential
  - Remember that $\mathbf{F}$ and $\mathbf{a}$ (force and acceleration) are **vectors**, while m (mass) is a **scalar**
- When a particle is in **equilibrium**, the vector sum of the forces on it is equal to zero

What is vector notation?

- There are **two vector notations** used in A level mathematics:
  - **Column vectors**: This is one number written above the other enclosed in brackets,

    e.g. The (column) vector meaning **3 units** in the **positive horizontal** (x) direction (i.e., **right**) and **2 units**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$\begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

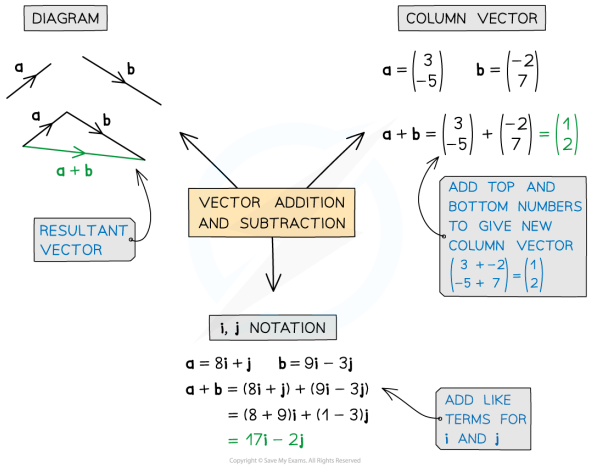
- **i** and **j** notation: **i** and **j** are **unit vectors** (they have **magnitude 1**) in the positive horizontal and positive vertical directions respectively

  e.g. The vector (-4**i** + 3**j**) would mean **4 units** in the **negative horizontal** (*x*) direction (i.e. **left**) and **3 units** in the **positive vertical** (*y*) direction (i.e. **up**)

- As they are vectors, **i** and **j** are displayed in **bold** in textbooks and online but in handwriting they would be underlined (i and j)

How do I calculate a resultant vector?

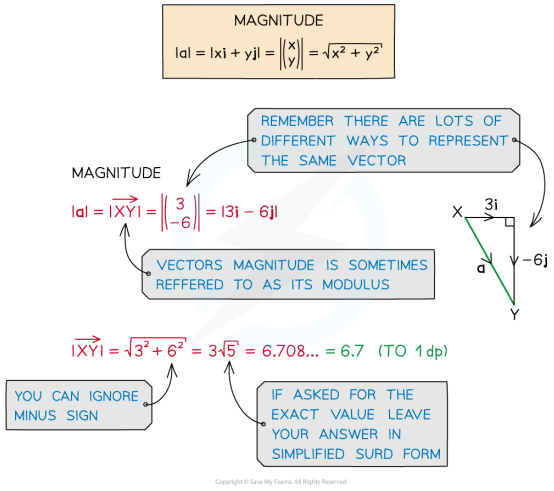
- Both **column vectors** and **i, j notation** can be used for calculating **resultant vectors**
- Adding vectors together gives the **resultant vector**
- This is the same when adding force vectors; the **resultant force** is simply the force vectors added together
- Forces in **equilibrium** have a resultant force equal to **zero**



How do I calculate the magnitude and direction of a vector?

- Pythagoras is used to find the magnitude of a vector.
- Trigonometry is used to find the **direction** of a vector
- You may be asked for the direction as a **bearing.** The **unit vector j** can be used to represent north and the unit vector **i** can be used to represent east.
  - In the case where $k > 0$
    - If a particle is moving north then its velocity will have a vector of k**j**
    - If a particle *A* is due south of another particle *B* then the displacement vector from *B* to *A* will have the form -k**j**
    - If a particle is moving east then its velocity will have a vector of k**i**
    - If a particle *A* is due west of another particle *B* then the displacement vector from *B* to *A* will have the form -k**i**

- If the position vector of a particle has equal **i** and **j** components then it is positioned due north - east of the origin
  - If a particle *A* is north – east of another particle *B* then the displacement vector from *B* to *A* will have equal **i** and **j** components



What does resolving vectors mean?

- **Resolving** a vector means writing it in **component** form (as **i** and **j** components)
- Given the **magnitude** and **direction** of a vector you can work out its components using trigonometry (SOHCAHTOA)

Worked Example

Two forces $\mathbf{F}_1$, $\mathbf{F}_2$ act on a particle P, $\mathbf{F}_1 = -7\mathbf{i} + 3\mathbf{j}$ and $\mathbf{F}_2 = 4\mathbf{i} + 6\mathbf{j}$.

a) Find the magnitude of the resultant force **R** acting on the particle.

Leaving your answer as a simplified surd.

Answer:

b) Find the bearing of the resultant force **R**.

A third force $\mathbf{F_3} = s\mathbf{i} - t\mathbf{j}$ brings the particle into equilibrium.

c) Find s and t and state the force for $\mathbf{F_3}$ in terms of **i** and **j**.

Answer:

Examiner Tips and Tricks

- When working with vectors pay attention to accuracy, leaving magnitude in surd form or correct to 3 s.f.
- In your exam you can't write in **bold** so should underline your vector notation

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.